Sec. VI-E, (98)–(108): the onset under a perturbation, and the threshold

drop-in main text; the measured sizes live in Sec. VIII-E

Because the measured angular rate $y(t)$ can be regarded as the nominal response perturbed by the unmodelled forcing ρ, the minimiser selects $\hat{t_c}$ by minimising the two-segment cost over the record. Both segments enter, and both share the pre-onset level C:

$$J(t_c) = \int_{t_0}^{t_c} \big(y(t) - C\big)^2 dt + \int_{t_c}^{t_{\text{end}}} \Big(y(t) - \big(\hat{\omega}(t;t_c) + C\big)\Big)^2 dt. \tag{98}$$

Write the recorded rate as the true pre-onset level, the nominal response from the true onset, and a deviation:

$$y(t) = b + \hat{\omega}(t;t_c^*) + e_\omega(t), \qquad \hat{t_c} = t_c^* + \delta t_c. \tag{99}$$

Here b is the rate the vehicle actually holds before the excitation, zero under the onset conditions. The rate is taken from the navigation filter, whose delta-angle bias states remove the gyro bias, so b represents residual pre-excitation motion rather than sensor bias; C is its estimate, formed over the pre-onset segment. Note also that e_ω is anchored at the *true* onset t_c^*: once a run is recorded it is a fixed function of t, and it carries no dependence on the optimisation variable.

A first-order expansion of the estimated rate gives

$$\hat{\omega}\big(t;\hat{t_c}\big) \;\simeq\; \hat{\omega}\big(t;t_c^*\big) + \delta t_c\,\chi(t) \tag{100}$$

with the sensitivity

$$\chi(t) \triangleq \frac{\partial \hat{\omega}(t;t_c)}{\partial t_c}\bigg|_{t_c = t_c^*} = -\,C_1 C_2 \sinh C_2\big(t - t_c^*\big). \tag{101}$$

Two features of (98) make the differentiation clean, and both are worth stating because each looks like an omission otherwise.

First, t_c is the upper limit of one integral and the lower limit of the other, so the Leibniz rule produces two boundary terms. They cancel identically. The rise contributes $C_1(\cosh 0 - 1) = 0$ at $t = t_c$, so the two segments agree there and both terms equal $\big(y(t_c) - C\big)^2$ with opposite signs. Nothing is discarded, and the cancellation is exact rather than asymptotic.

Second, substituting (99) and (100), the post-onset residual is

$$y(t) - \hat{\omega}(t;\hat{t_c}) - C = e_\omega(t) - \delta t_c\,\chi(t) - (C - b), \tag{102}$$

so C enters only through the baseline *error* $C - b$, not through its own size. A large but correctly estimated pre-onset level is harmless.

Differentiating (98) with respect to δt_c — the same operator as $\partial/\partial t_c$ by (99), but the variable in which the linearised cost is an exact quadratic — and setting the result to zero gives, when the pre-onset fit is unbiased,

$$\frac{\partial J}{\partial\, \delta t_c} = -2 \int_{t_c}^{t_{\text{end}}} \big(e_\omega(t) - \delta t_c\,\chi(t)\big)\chi(t)\,dt = 0, \tag{103}$$

which implies

$$\delta t_c = \frac{\langle e_\omega, \chi\rangle}{\|\chi\|^2}. \tag{104}$$

Here $\langle a, b\rangle$ denotes the L^2 inner product over the fitting window $[t_c^*, t_{\text{end}}]$, i.e. $\langle a, b\rangle \triangleq \int_{t_c^*}^{t_{\text{end}}} a(t)b(t)\,dt$, and $\|b\|^2 \triangleq \langle b, b\rangle$. The shift δt_c is well defined because the maximum-angle event occurs after the onset, $\tau_{\text{end}} > 0$, so $\sinh C_2\tau$ does not vanish identically and $\|\chi\|^2 > 0$.

The baseline error, retained. Keeping the last term of (102) rather than assuming it away replaces (104) by

$$\delta t_{\mathrm{c}} = \frac{\langle e_\omega, \chi\rangle}{\|\chi\|^2} \; - \; (C-b)\,\frac{\langle 1, \chi\rangle}{\|\chi\|^2}, \tag{105}$$

and the second term does not vanish: χ is of one sign, so $\langle 1, \chi\rangle = -C_1(\cosh x - 1)$. Evaluating it with $\|\chi\|^2 = C_1^2 C_2\big[\frac{1}{4}\sinh 2x - \frac{1}{2}x\big]$ and carrying it through $\Delta M_{\mathrm{crit}} = \dot{M}\,\delta t_{\mathrm{c}}$, the amplitude C_1 cancels as it does elsewhere and

$$\Delta M_{\mathrm{crit},C} = W z_{CoM}\,(C-b)\,\frac{\cosh x - 1}{C_2\big[\frac{1}{4}\sinh 2x - \frac{1}{2}x\big]}. \tag{106}$$

Over the operating box of Sec. VI-D the gain of (106) runs from 0.042 N m per rad/s at the slowest ramp to 0.351 at the fastest, the fast end being worse because the window is shorter. What the fitted baselines actually are, and how far the identified offset moves when C is pinned to zero instead of fitted, are measurements and are reported with the realised budget in Sec. VIII-E.

From the onset shift to the threshold. Substituting the Duhamel integral (88) for e_ω in (104) and exchanging the order of integration — Fubini, both integrands being continuous on a bounded domain, the inner limits changing from $s \in [0, \tau]$ to $\tau \in [s, \tau_{\mathrm{end}}]$ — turns the correlation into an average of ρ against a fixed weight:

$$\Delta M_{\mathrm{crit},e} = -\int_0^{\tau_{\mathrm{end}}} \rho(s)\, w(s)\, ds, \quad w(s) = \frac{W z_{CoM} \displaystyle\int_s^{\tau_{\mathrm{end}}} \sinh(C_2\tau)\cosh\big(C_2(\tau - s)\big)\, d\tau}{J_P C_2\, \big\|\sinh(C_2\tau)\big\|^2}. \tag{107}$$

Three properties of w close the argument. It is non-negative, being an integral of hyperbolic functions of non-negative arguments. It integrates to one *exactly*: the same exchange over the triangle $0 \le s \le \tau \le \tau_{\mathrm{end}}$ reproduces in the numerator the very norm that sits in the denominator, leaving $W z_{CoM}/(J_P C_2^2) = 1$ by (20). And it is non-increasing in s, since both the integration range and the cosh factor shrink as s grows.

The first two properties alone would give $|\Delta M_{\mathrm{crit},e}| \le \rho_{\max}$. The third admits Chebyshev's inequality a second time — a forcing that rises against a weight that falls, the same pairing as before — and replaces that maximum by the window average already bounded at (95):

$$\big|\Delta M_{\mathrm{crit},e}\big| \;\le\; \bar{\rho} \;\le\; \tfrac{1}{7}\,\rho_{\varphi,\max} + \tfrac{1}{5}\,\rho_{GE,\max}. \tag{108}$$

Note that $\bar{\rho}$ is the average defined at (95), and the right-hand side bounds it rather than defining it. The constants here are the reduction factors themselves, not the factors multiplied by a kernel gain, because the onset is located by a *normalised* correlation: $\int w = 1$ leaves nothing over.

That the two channels rise across the window is what makes the Chebyshev step available, and it is also the pessimistic case rather than a convenient one. Expanding the kernel of (88) as $\cosh C_2(\tau - s) = \cosh C_2\tau \cosh C_2 s - \sinh C_2\tau \sinh C_2 s$, the part of the deviation that displaces the onset is weighted by $\sinh C_2 s$, which vanishes at $s = 0$. A perturbation acting at the onset therefore moves the identified threshold not at all — it changes the amplitude instead, and the amplitude is pinned by $C_1 = K\dot{M}$, so it shows in the residual rather than in the answer. Only forcing that acts *late* is weighted appreciably, and $\rho_\varphi \propto \varphi^2$ and $\rho_{GE} \propto \tau\varphi$ do exactly that. The bound is thus stated for the arrangement of ρ least favourable to it.

The constants differ between the two halves of (97) because the two questions do. The first asks how far the fitted rate curve is displaced, so the kernel integral Ψ multiplies $\bar{\rho}$ and the products climb to $\frac{1}{2}$ and 1; the second asks how far the *minimiser* moves, and by (107) that is a normalised correlation with no kernel gain left over, so $\bar{\rho}$ enters bare and the constants are the reduction factors themselves. The threshold bound is the stronger of the two by about a factor of four, which is why it, and not the relative rate bound, is what the offset budget inherits. A reviewer can evaluate either from numbers already in the paper.

What (108) covers. Only the displacement driven by the forcing the estimator cannot absorb. Two further contributions sit outside it and are measured rather than bounded a priori. The baseline error enters through (106), so that $|\Delta M_{\mathrm{crit}}| \le \bar{\rho} + |\Delta M_{\mathrm{crit},C}|$. And the applied moment departs from the linear ramp the model assumes: because $\dot{M}$ is the least-squares slope of the *measured* moment over the window, the constant and linear parts of any tracking error are removed by the normal equations, but the non-affine remainder survives and pairs with the non-affine part of w. Both are reported with the realised budget in Sec. VIII-E, alongside the ground-contact geometry that dominates all three.